

AI ASSISTENCE CODING- 6.5

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Experiment 6: AI-Based Code Completion: Working with suggestions for classes, loops, conditionals Week3 - Friday

LO1. Use AI-based code completion tools to generate Python code involving classes, loops, and conditionals.

LO2. Interpret and explain AI-generated code line-by-line.

LO3. Identify errors, inefficiencies, or logical flaws in AI-suggested implementations.

LO4. Optimize AI-generated code for better readability and performance.

LO5. Demonstrate ethical and responsible use of AI tools in coding tasks.

Task Description #1

(AI-Based Code Completion for Conditional Eligibility Check)

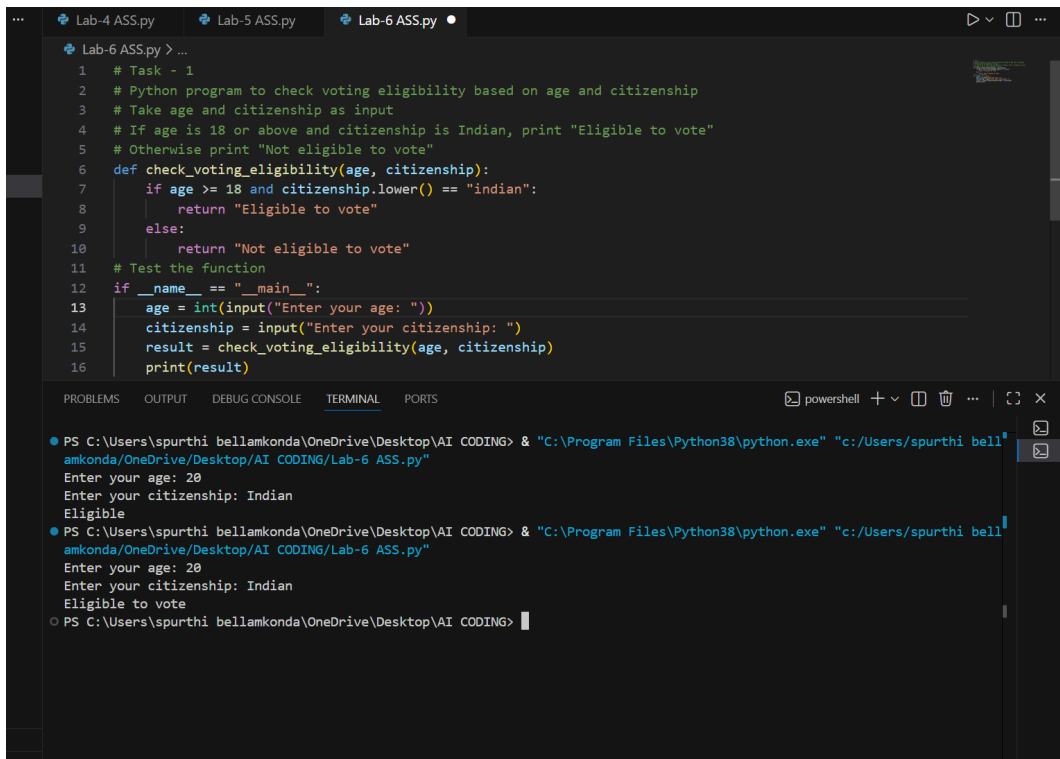
Task: Use an AI tool to generate eligibility logic.

Prompt:

"Generate Python code to check voting eligibility based on age and citizenship."

Expected Output:

- AI-generated conditional logic.
- Correct eligibility decisions.
- Explanation of conditions.



The image shows a Visual Studio Code editor window with three tabs: 'Lab-4 ASS.py', 'Lab-5 ASS.py', and 'Lab-6 ASS.py'. The 'Lab-6 ASS.py' tab is active, displaying a Python script. The script is a program to check voting eligibility based on age and citizenship. It includes a function `check_voting_eligibility` that takes `age` and `citizenship` as arguments. The function checks if the age is 18 or above and if the citizenship is 'indian' (case-insensitive). If both conditions are met, it returns 'Eligible to vote'; otherwise, it returns 'Not eligible to vote'. The script also includes a `__main__` block that prompts the user for age and citizenship, calls the function, and prints the result.

```
1 # Task - 1
2 # Python program to check voting eligibility based on age and citizenship
3 # Take age and citizenship as input
4 # If age is 18 or above and citizenship is Indian, print "Eligible to vote"
5 # Otherwise print "Not eligible to vote"
6 def check_voting_eligibility(age, citizenship):
7     if age >= 18 and citizenship.lower() == "indian":
8         return "Eligible to vote"
9     else:
10        return "Not eligible to vote"
11 # Test the function
12 if __name__ == "__main__":
13     age = int(input("Enter your age: "))
14     citizenship = input("Enter your citizenship: ")
15     result = check_voting_eligibility(age, citizenship)
16     print(result)
```

Below the editor, the 'TERMINAL' panel shows the execution of the script. It displays the command to run the Python file and the user's input: age 20 and citizenship Indian. The output is 'Eligible to vote'.

```
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda/OneDrive/Desktop/AI CODING/Lab-6 ASS.py"
Enter your age: 20
Enter your citizenship: Indian
Eligible
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda/OneDrive/Desktop/AI CODING/Lab-6 ASS.py"
Enter your age: 20
Enter your citizenship: Indian
Eligible to vote
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING>
```

Task Description #2

(AI-Based Code Completion for Loop-Based String Processing)

Task: Use an AI tool to process strings using loops.

Prompt:

"Generate Python code to count vowels and consonants in a string using a loop."

Expected Output:

- AI-generated string processing logic.
- Correct counts.
- Output verification

The image shows a VS Code editor window with a Python file named 'Lab-6 ASS.py'. The code is a Python program to count vowels and consonants in a string using a loop. It includes comments, a function definition, and an example usage section. The terminal output shows the program being executed, with prompts for age, citizenship, and a string, followed by the output 'Vowels: 2, Consonants: 5'.

```
18
19 # Task 2
20 # Python program to count vowels and consonants in a string using a loop
21 # Ignore spaces and non-alphabet characters
22 # Print the total number of vowels and consonants
23 def count_vowels_consonants(s):
24     (variable) vowel_count: Literal[0]
25     vowel_count = 0
26     consonant_count = 0
27     for char in s:
28         if char.isalpha():
29             if char in vowels:
30                 vowel_count += 1
31             else:
32                 consonant_count += 1
33     print(f"Vowels: {vowel_count}, Consonants: {consonant_count}")
34 # Example usage
35 input_string = input("Enter a string: ")
36 count_vowels_consonants(input_string)
```

Terminal Output:

```
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda/OneDrive/Desktop/AI CODING/Lab-6 ASS.py"
Enter your age: 20
Enter your citizenship: Indian
Eligible to vote
Enter a string: Spurthi
Vowels: 2, Consonants: 5
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING>
```

Task Description #3

(AI-Assisted Code Completion Reflection

Task)

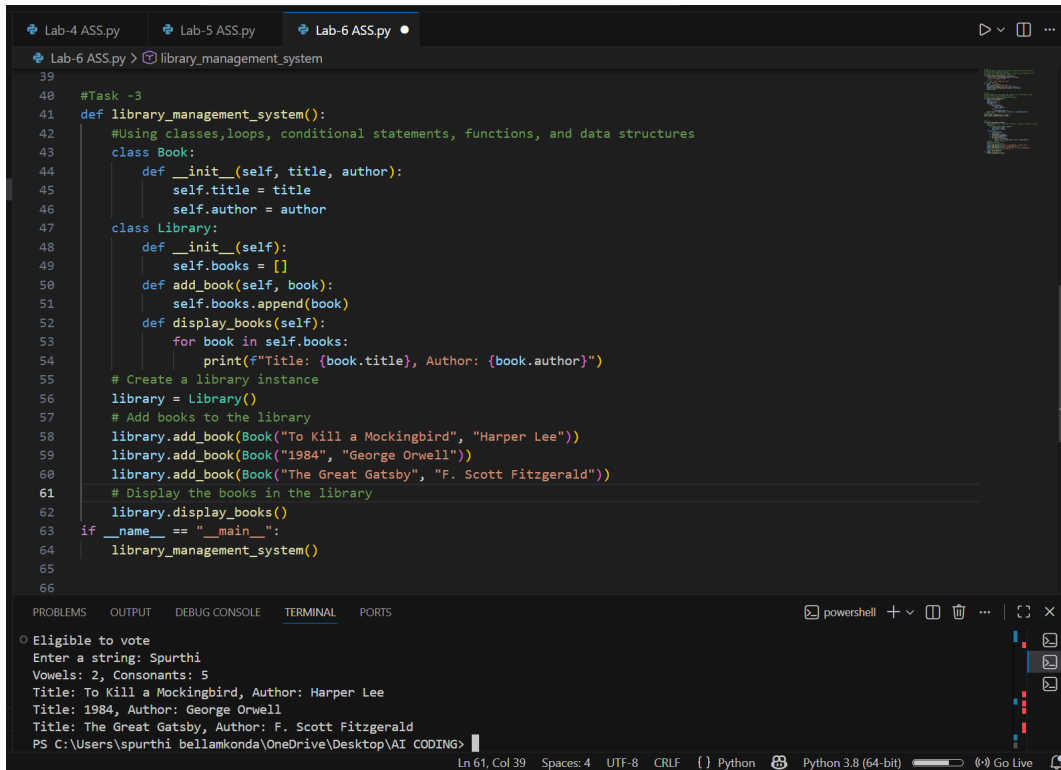
Task: Use an AI tool to generate a complete program using classes, loops, and conditionals.

Prompt:

"Generate a Python program for a library management system using classes, loops, and conditional statements."

Expected Output:

- Complete AI-generated program.
- Review of AI suggestions quality.
- Short reflection on AI-assisted coding experience.



```
39
40 #Task -3
41 def library_management_system():
42     #Using classes, loops, conditional statements, functions, and data structures
43     class Book:
44         def __init__(self, title, author):
45             self.title = title
46             self.author = author
47     class Library:
48         def __init__(self):
49             self.books = []
50         def add_book(self, book):
51             self.books.append(book)
52         def display_books(self):
53             for book in self.books:
54                 print(f"Title: {book.title}, Author: {book.author}")
55     # Create a library instance
56     library = Library()
57     # Add books to the library
58     library.add_book(Book("To Kill a Mockingbird", "Harper Lee"))
59     library.add_book(Book("1984", "George Orwell"))
60     library.add_book(Book("The Great Gatsby", "F. Scott Fitzgerald"))
61     # Display the books in the library
62     library.display_books()
63 if __name__ == "__main__":
64     library_management_system()
65
66
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
o Eligible to vote
Enter a string: Spurthi
Vowels: 2, Consonants: 5
Title: To Kill a Mockingbird, Author: Harper Lee
Title: 1984, Author: George Orwell
Title: The Great Gatsby, Author: F. Scott Fitzgerald
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING>
```

Ln 61, Col 39 Spaces: 4 UTF-8 CRLF Python Python 3.8 (64-bit) Go Live

Task Description #4

(AI-Assisted Code Completion for Class-Based Attendance System)

Task: Use an AI tool to generate an attendance management class.

Prompt: "Generate a Python class to mark and display student attendance using loops."

Expected Output:

- AI-generated attendance logic.
- Correct display of attendance.
- Test cases.

```
Lab-6 ASS.py > ...
67 # Task - 4
68 # Task 4: Student Attendance System
69 # Create a simple Attendance class
70 # Use loop to input student names and P/A status
71 # Store attendance in a dictionary
72 # Display present and absent students
73 class Attendance:
74     def __init__(self):
75         self.attendance_record = {}
76     def mark_attendance(self, name, status):
77         self.attendance_record[name] = status
78     def display_attendance(self):
79         present_students = [name for name, status in self.attendance_record.items() if status == 'P']
80         absent_students = [name for name, status in self.attendance_record.items() if status == 'A']
81         print(f"Present Students: {' '.join(present_students)}")
82         print(f"Absent Students: {' '.join(absent_students)}")
83 # Example usage
84 if __name__ == "__main__":
85     attendance = Attendance()
86     num_students = int(input("Enter the number of students: "))
87     for _ in range(num_students):
88         name = input("Enter student name: ")
89         status = input("Enter attendance status (P/A): ")
90         attendance.mark_attendance(name, status)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Enter a string: Spurthi
Vowels: 2, Consonants: 5
Title: To Kill a Mockingbird, Author: Harper Lee
Title: 1984, Author: George Orwell
Title: The Great Gatsby, Author: F. Scott Fitzgerald
Enter the number of students: 3
Enter student name: Alice
Enter attendance status (P/A): P
Enter student name: Bob
Enter attendance status (P/A): P
Enter student name: Charlie
Ln 93, Col 1 Spaces: 4 UTF-8 CRLF { } Python Python 3.8 (64-bit) Go Live
```

Task Description #5

(AI-Based Code Completion for Conditional Menu Navigation)

Task: Use an AI tool to complete a navigation menu.

Prompt: "Generate a Python program using loops and conditionals to simulate an ATM menu."

Expected Output:

- AI-generated menu logic.
- Correct option handling.
- Output verification.

```

Lab-4 ASS.py  Lab-5 ASS.py  Lab-6 ASS.py •
Lab-6 ASS.py > ...
94 #Task 5
95 # ATM Menu Simulation
96 # Initialize balance
97 # Use while loop to display menu
98 # Options: check balance, deposit, withdraw, exit
99 # Use conditionals to handle each option
100 balance = 1000 # Initial balance
101 while True:
102     print("\nATM Menu:")
103     print("1. Check Balance")
104     print("2. Deposit Money")
105     print("3. Withdraw Money")
106     print("4. Exit")
107     choice = input("Enter your choice (1-4): ")
108     if choice == "1":
109         print(f"Your current balance is: ${balance}")
110     elif choice == "2":
111         amount = float(input("Enter amount to deposit: "))
112         balance += amount
113         print(f"Deposited ${amount}. New balance is: ${balance}")
114     elif choice == "3":
115         amount = float(input("Enter amount to withdraw: "))
116         if amount <= balance:
117             balance -= amount

```

```

Lab-4 ASS.py  Lab-5 ASS.py X  Lab-6 ASS.py •
Lab-6 ASS.py > ...
111         amount = float(input("Enter amount to deposit: "))
112         balance += amount
113         print(f"Deposited ${amount}. New balance is: ${balance}")
114     elif choice == "3":
115         amount = float(input("Enter amount to withdraw: "))
116         if amount <= balance:
117             balance -= amount
118             print(f"Withdrew ${amount}. New balance is: ${balance}")
119         else:
120             print("Insufficient funds.")
121     elif choice == "4":
122         print("Thank you for using the ATM.")
123         break
124     else:
125         print("Invalid choice. Please try again.")

```

PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda/Desktop/AI CODING/Lab-6 ASS.py"

```

ATM Menu:
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice (1-4): 3
Enter amount to withdraw: 300
Withdrew $300.0. New balance is: $1200.0

```

```

ATM Menu:
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit

```